Impact of Hurricane Sandy on Residential Property Value

Team: Langley, Bryan, Gina, Aritra

Overview

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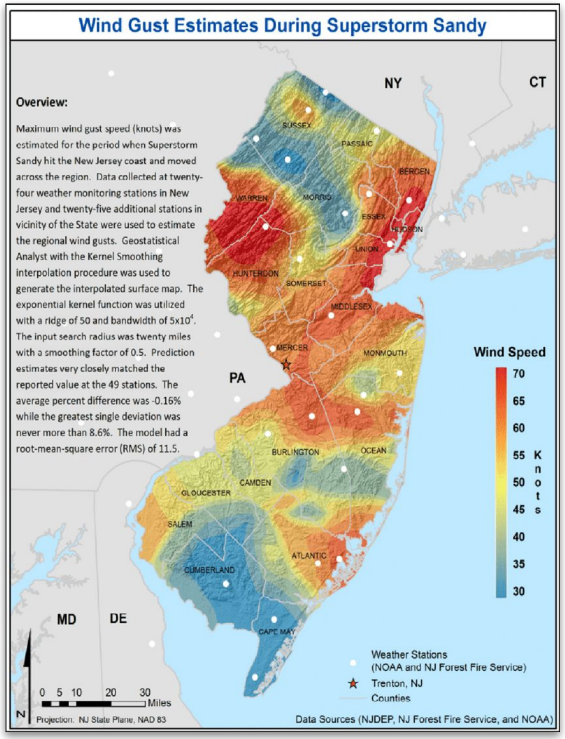
- Data Sources
- Data Pre-processing
- Methodology & Experiment
- Results
- Robustness Checks
- Conclusion

Introduction

- Impact of Hurricane Sandy on sale assessment values in Atlantic County, NJ.
- Hurricane Sandy Overview:
  - Struck northeastern U.S. in October 2012; NJ hardest hit.
  - $70 billion in damages; 340,000+ homes affected in NJ (FEMA).
  - Significant property damage, tourism decline.
- Research Methodology:
  - Difference-in-Differences (DiD) analysis comparing Atlantic County (treatment) and Sussex County (control).
  - Validated with balance tests and a falsification test
- Significance:
  - Insights for policymakers and homeowners on housing market resilience post-disaster.
  - Supports planning for future disaster recovery and insurance policies.

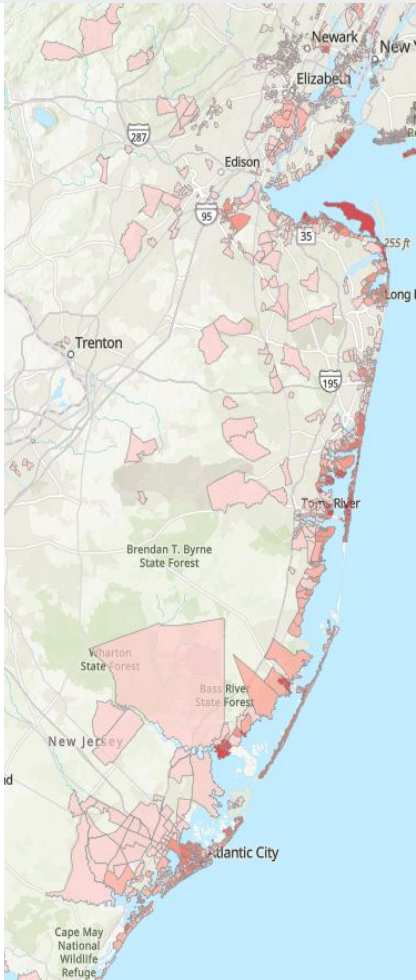


Literature Review

- Natural disasters have big impact - Floods 40% of natural disasters worldwide with U.S. damage totalling over $260 billion*
- Climate change and the increased value of property at risk have increased costs
- **Areas Studied in Literature:** Florida, Georgia, Missouri, New York, Texas, and various hurricanes across U.S.

Models Used:

- Difference-in-Differences (DiD)/Staggered DiD: Compares pre/post disaster outcomes between treated/control
- Hedonic Pricing: Decomposes property values into components like location and neighborhood features. Measures environmental premium.
- Spatial Econometric Models (SARAR): Captures spatial dependencies (nearby properties influence each other)
- Panel Event-Study Designs: Tracks dynamic impacts, controlling for pre-trends and socioeconomic factors.
- Dynamic Panel Models: Examines temporal effects using metrics like a hurricane destruction index.
- Repeat Sales Models: Tracks price changes over time for the same property to isolate temporal effects.
- Fixed Effects Models: Controls for unobserved, time-invariant characteristics at property or regional levels.



*2009 constant dollars, between 1955 to 2009

Literature Review (cont'd)

Literature Results: Analyzes immediate and lasting effects on property values with mixed results

- *Atreya et al. (2013)*: 100-year floodplain saw **25–44% price drops** post-Tropical Storm Alberto, but effects **dissipated within 4–9 years**
- *Kousky (2010)*: Great Midwest Flood of 1993 showed **no significant change** in 100-year floodplain prices but a **2–5% decline** in 500-year floodplain properties.
- *Murphy & Strobl (2009):* **Temporary 3–4% price increases** due to supply shocks.
- *Zivin et al. (2020):* **10% transitory price increases** in Florida post-hurricane but demographic shifts toward higher-income buyers.
- *Ortega & Taspinar (2018):* **Persistent 9% price reduction** in flood-prone NYC neighborhoods post-Hurricane Sandy.

Our Study

- Our intensive DiD event study examines how Hurricane Sandy affected property-level sales assessment values in NJ, using FEMA damage intensity estimates
- Compares more affected, coastal Atlantic County to less affected, mountainous Sussex County
- Explores coastal property resilience and contributes to literature to support disaster resilience planning and inform public policy planning

Data - Sources

- Data was collected on residential properties from the New Jersey Property Tax System, known as the MOD-IV Historical Database
  - Variables included: County, Municipality, Sales Assessment Value, Address, Date of Sale, Residential Status

- The demographic data for Atlantic and Sussex County, NJ was taken from the U.S. Census Bureau, using the ACS (American Community Survey) data for 1 year estimates in both 2010 and 2011.

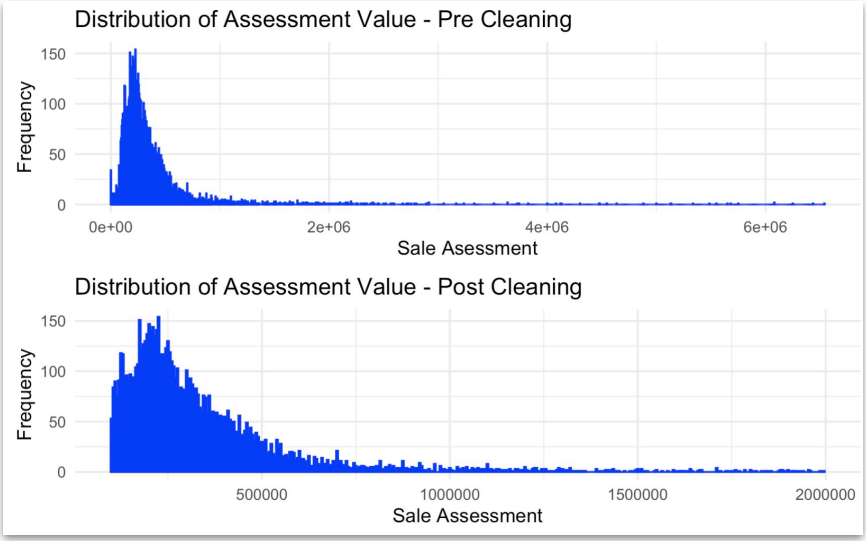
- Data was pulled from the Office of Policy Development and Research at the Department of Housing and Urban Development (HUD) to understand the percentage amount of damage done to various areas of NJ from Hurricane Sandy
  - Contains map of Sandy damage estimates by block group based on FEMA Individual Assistance (IA) Registrant Inspection data
  - Measures height and extent of flooding, where applicable
  - For property without flooding, HUD estimates minor/major/severe damage

N.J. MOD IV Historical Database

U.S. Department of Housing and Urban Development

Data - Cleaning

- High number of zero values
- Focused on Residential properties - helped remove some of these zero values which were encompassed of parking lots or vacant land.
- Noticed that some entries still had unusually low or zero sale values.
- Excluded properties with sale valuation of below or equal to $100,000 or below or equal to $2 million.
- These high-value transactions were considered outliers and were excluded to prevent them from skewing the average residential sale values.

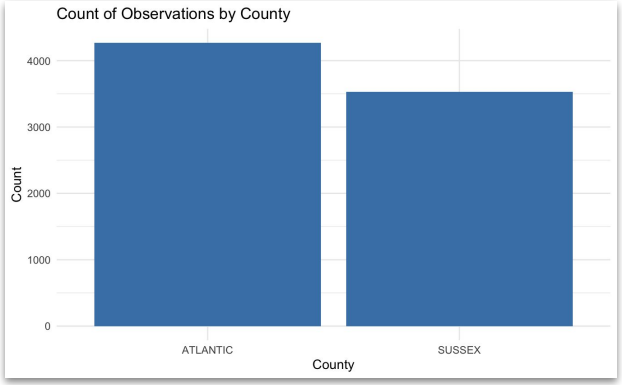


Sale Assessment Summary Stats Before and After Capping

	Mean	Median	Minimum	Maximum	Skew	SD
SaleAssessmentPre	357,129	261,000	0	6,540,600	6.86	410,426
SaleAssessmentPost	340,298	269,550	100,000	1,996,800	2.85	247,503

Exploratory Data Analysis

- A slight difference in the number of observations was noted between Atlantic County and Sussex County
- This is likely due to differences in population size, which will be highlighted in the demographics table
- Overview of the sale assessment statistics for both counties - allowing for a comparison of housing values before the experiment
    - Notably, the skewness for Atlantic County decreased from 5.5 to 2.3 after capping the values
    - These higher values should not be interpreted as indicative of extreme differences, it is simply highlighting the higher assessment value of housing in Atlantic County from 2010-1014



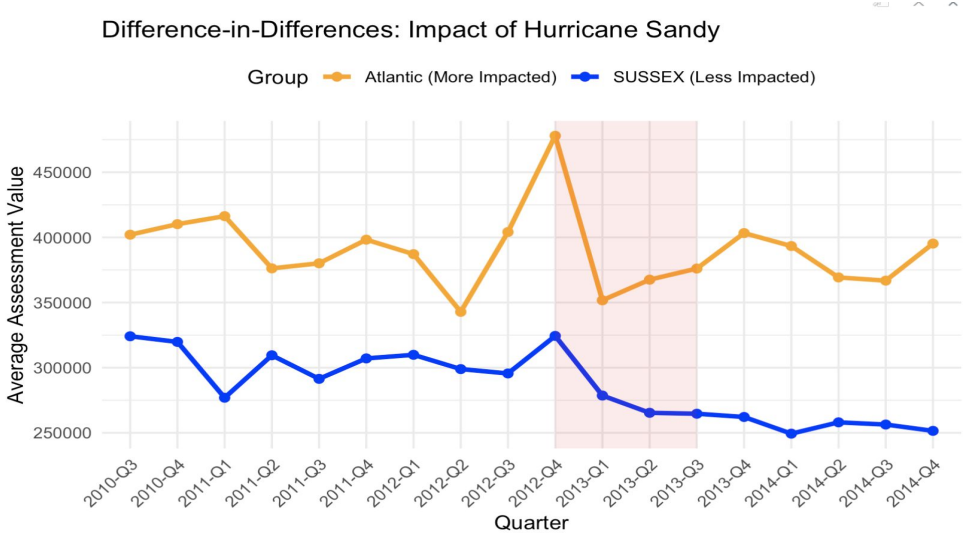
Summary of Sale Assessment Value By Group - Post Cleaning

CountyName	Mean	Median	SD	Min	Max	Skew
ATLANTIC	388735.8	302800	299075.2	1e+05	1996800	2.337991
SUSSEX	281763.9	245700	144728.8	1e+05	1784000	2.551587

Exploratory Data Analysis

When selecting two counties for model comparison, three key principles were applied to ensure a robust analysis:

1. **Sufficient Pre- and Post-Intervention Data**:
   a. At least 8 quarters of data before and after the natural disaster in October 2012 were required to establish reliable baseline trends and capture long-term effects.
2. **Parallel Trends Before the Disaster**:
   a. The counties needed to exhibit similar trends in property prices before the disaster.
3. **Post-Intervention Divergence**:
   a. A clear change in property price trends post-disaster was necessary, with the treatment county showing a noticeable impact while the control county remained relatively unaffected.
   b. Post hurricane impact, there is a divergence in the parallel trends, Atlantic county's price valuation rises while Sussex remains steady / decreases slightly.



Methodology and Experiment

Difference-in-Differences (DiD) Technique:
Difference-in-Differences is a powerful quasi-experimental research design that enables researchers to estimate the causal effect of a specific intervention or event—in this case, hurricane impact—by comparing changes in economic outcomes between treatment and control groups over time.

The underlying assumptions of the DiD approach are:

1. **Parallel Trends**: Prior to the intervention, the outcome variable trends for both groups must be similar. This ensures that observed differences post-treatment can be attributed to the intervention.
2. **No Spillover Effects**: The treatment should only affect the treated group, with no indirect effects on the control group.
3. **Consistency**: The treatment and control groups are consistently observed, with no unobserved heterogeneity that correlates with both the treatment and the outcome.

The estimating equation for our model is:

$$\text{Sale Assessment}_{it} = \beta_0 + \beta_1 \text{Impacted}_i + \beta_2 \text{Time}_t + \beta_3 (\text{Impacted}_i \cdot \text{Time}_t) + \epsilon_{it}$$

Methodology and Experiment (cont'd)

The Difference-in-Differences (DiD) approach utilized a linear regression model

- **Sale Assessment:** Assessed value of the residential property at the time of sale in Sussex and Atlantic counties.
- **Time**: Indicates whether the event (hurricane) has occurred
  - 0 = before the event
  - 1 = after the event.
  - Helps the model distinguish between the periods before and after the hurricane.
- **Treated / Impacted:** Reflects the intensity of the hurricane's impact on each county. The intensity <u>was calculated as the average impact percentages of cities within each county</u>, as reported by FEMA.
  - **Atlantic County**: Impact percentages ranged from 47.1% to 100%, with an average of approximately 73%.
  - **Sussex County**: Impact percentages ranged from 0.5% to 9.8%, with an average of approximately 5%.
- **Interaction (Time x Treated)**: Captures the change in the dependent variable (residential sale assessment) attributable to the intervention, measuring the difference in the outcome for the treated group compared to the control group, adjusted for the intensity of the hurricane's impact after the event.

	Dependent variable:
	sale_assessment
Treated	125,868.900***
	(12,321.010)
Time	-39,944.720***
	(8,867.074)
Interaction	53,759.800***
	(16,323.120)
Constant	297,237.900***
	(6,776.968)
Observations	7,798
R2	0.049
Adjusted R2	0.048
Residual Std. Error	241,437.000 (df = 7794)
F Statistic	133.240*** (df = 3; 7794)
Note:	*p<0.1; **p<0.05; ***p<0.01

Results

Econometric Model Results

- Interaction effect: For counties impacted by the higher treatment intensity, for every unit increase, sale values increased by $53,760 more in the post-hurricane period compared to the pre-hurricane period.
- This is the key difference in difference effect, indicating that higher intensity hurricane impacts are associated with higher sale values post hurricane relative to less impacted properties.
- Coefficient is statistically significant ***

Model Characteristics

- Adjusted R^2 of 0.049 indicates complex underlying market dynamics
- Statistically significant F-statistic (133.2) confirms model's predictive validity
- Residual standard error of $241,437 reflects market variability

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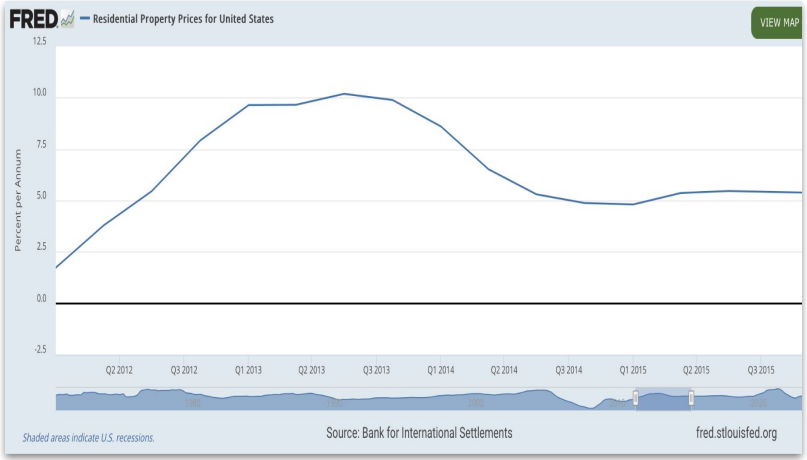
Results (cont'd)

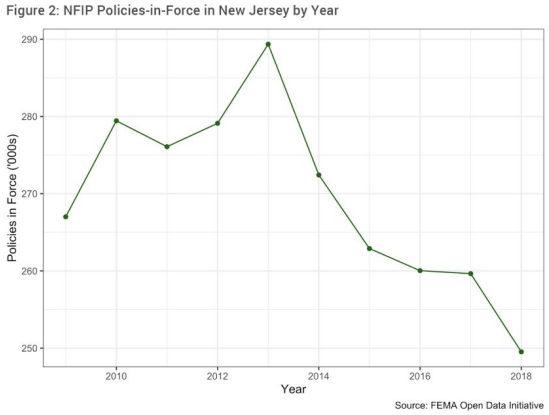
Property Value Dynamics

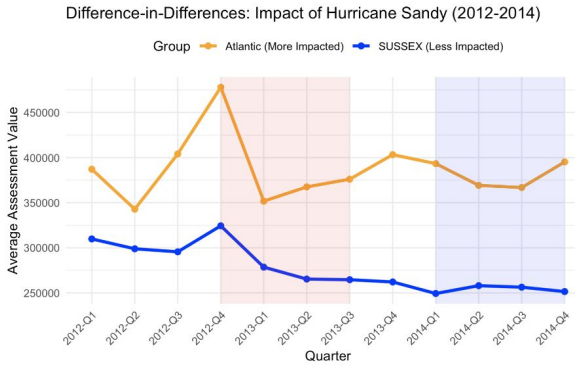
- Initial residential property valuations dropped immediately after Hurricane Sandy
- Property prices recovered within 3 quarters, reflecting limited supply and the inherent valuable location characteristics that would hold post-disaster.
- Atlantic County deviated from U.S Residential Property Price trends

Market Recovery Mechanisms

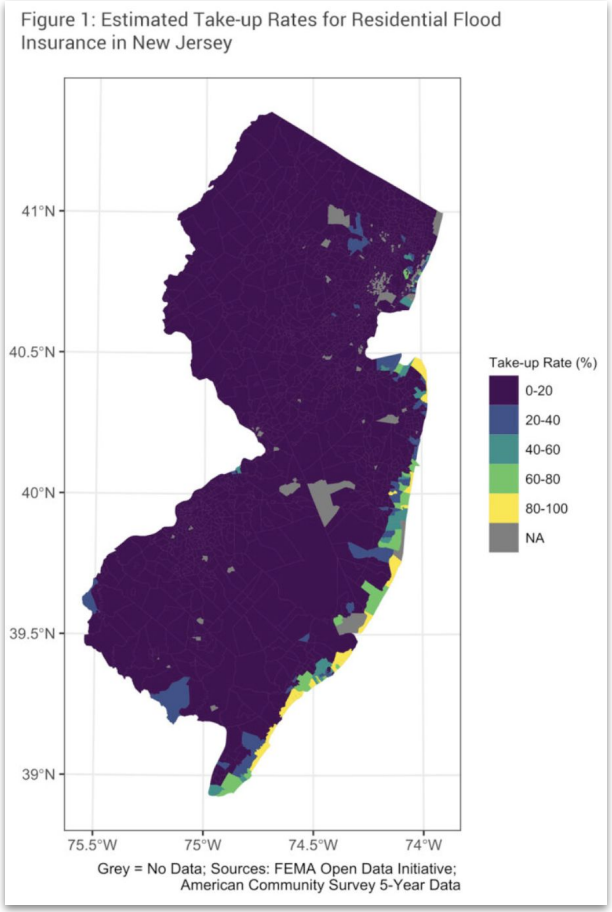
- National Flood Insurance Program (NFIP) likely played a crucial role in market stabilization
- Federal aid and insurance payouts supported property value recovery







Results (cont'd)



Balance Test

Balance Test

- Ensuring the characteristics of both counties are comparable pre hurricane
- Computed p values for demographics in both counties
- **Those that were statistically significant:**
  - Population Density: P-Value of 0.00
  - Household Income: P-Value of 0.01
- **Key Significance Test**
  - Median Assessment Values: P-Value of 0.08
  - Not significantly different (Pre Hurricane Sandy)
- This comparability supports the robustness of the difference-in-differences approach for assessing the impact of Hurricane Sandy on housing assessment values.

Balance Table

Variable	Mean (Sussex)	Mean (Atlantic)	SD (Sussex)	SD (Atlantic)	Difference in Means	P-Value
PopulationDensity	148878.00	274511.50	510.53	245.37	-125633.50	0.00
Male	0.50	0.48	0.01	0.00	0.01	0.17
Female	0.50	0.52	0.01	0.00	-0.01	0.17
MedianAge	41.90	40.25	0.57	0.21	1.65	0.12
CollegeDegree	0.44	0.48	0.06	0.02	-0.05	0.42
HouseholdIncome	83977.00	51700.00	195.16	1231.78	32277.00	0.01
UnemploymentRate	0.11	0.14	0.00	0.02	-0.02	0.26
HouseSize	2.68	2.66	0.01	0.01	0.02	0.29
OwenedHomes	0.84	0.70	0.01	0.05	0.14	0.14
MarriedFamilyOwners	0.61	0.45	0.00	0.02	0.16	0.06
AssessmentValue	277425.00	316900.00	8874.19	2687.01	-39475.00	0.08

Limitations and Future Research

Limitations & Future Research

- Constrained by publicly available property pricing data
- Many searchable databases but limited data sets
- Recommended future studies utilize comprehensive commercial databases
- Potential for more granular analysis with expanded data access

Conclusion

In conclusion, Hurricane Sandy's impact on Atlantic County's housing market was complex and nuanced. While the initial shock of the disaster led to temporary price declines, the long-term effects were surprisingly positive. Robust recovery efforts, federal aid, and insurance payouts played a crucial role in stabilizing and even boosting property values.

This study underscores the importance of disaster preparedness, effective recovery strategies, and strong insurance mechanisms in mitigating the economic consequences of natural disasters. By learning from such experiences, we can better equip communities to withstand future challenges and ensure their long-term resilience.

Thank you!

Q&A